

Colorectal Cancer Atlas Re-Analysis: Harmony vs scVI Integration Benchmark

Single-cell RNA-seq of MMRd and MMRp colorectal tumors (Pelka et al., 2021)

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Summary

We re-analyzed the publicly available colorectal cancer (CRC) single-cell atlas from Pelka et al. (Cell, 2021) to benchmark two widely used batch-integration strategies on a balanced cohort of mismatch-repair-deficient (MMRd) and mismatch-repair-proficient (MMRp) tumors. Eight donors (4 MMRd + 4 MMRp), tumor samples only, were selected from the CELLxGENE-curated atlas, yielding 56,022 cells after QC. We applied **Harmony** (linear PCA-based correction) and **scVI** (deep generative model on raw counts) using donor as the batch covariate, and quantified both batch removal and biological signal preservation. The two methods occupy distinct points on the integration trade-off curve: Harmony delivered the strongest batch mixing (donor mixing entropy 1.82 vs 1.04 baseline, +75%), while scVI gave the cleanest lineage structure (silhouette 0.293 vs 0.198 baseline, +48%). Compositional analysis recovered the canonical MMRd immune signature: elevated T/NK/ILC infiltration in MMRd tumors versus plasma-cell dominance in MMRp tumors, consistent with the immunotherapy-responsive phenotype of microsatellite-unstable CRC.

Dataset and cohort selection

The source data is the CELLxGENE-hosted release of the Pelka 2021 CRC atlas (collection ID 4d82bd8e-8827-410b-8fc7-685b7dd92585, DOI 10.1016/j.cell.2021.08.003). The full atlas contains 370,115 cells across 62 donors and 7 lineage categories. We restricted analysis to tumor-derived cells (**disease = colon adenocarcinoma**) and selected the top four donors by cell count within each MMR status group, balancing the comparison and keeping runtime tractable on a workstation-class CPU.

Selected donors:

- MMRd (MSI-like): C123, C130, C139, C170
- MMRp (MSS-like): C124, C129, C162, C171

After per-cell QC (`n_genes` $\geq$ 500, `pct_mt` $\leq$ 25) and gene filtering (`min_cells` $\geq$ 10), the working object contained **56,022 cells x 23,415 genes**. Donor-aware highly variable gene selection (Scanpy `seurat_v3` flavor on raw counts) retained 3,000 genes for downstream embedding.

Methods

Normalization. Counts were normalized per cell to 10,000 and log1p-transformed. The raw count matrix was retained in a separate layer for scVI training.

Baseline embedding. PCA on scaled HVG expression (50 components) followed by nearest-neighbor graph ($k = 15, 30$ PCs) and UMAP.

Harmony. harmonypy v0.0.10+ called directly on the baseline PCA. Default theta and lambda; 20 maximum iterations; convergence reached at iteration 5.

scVI. Variational autoencoder trained on raw counts with `donor_id` as the batch covariate. Architecture: 10-dimensional latent space, 2 hidden layers, 128 units per layer, negative binomial likelihood. Training: up to 50 epochs, batch size 512, learning rate 1e-3, early stopping with patience 8, CPU backend.

Clustering. Leiden (igraph backend, resolution 0.6) on the integrated neighbor graph for each method.

Integration metrics. Two complementary scores were computed on 5,000 randomly sampled cells (seed fixed):

- *Donor mixing entropy:* Shannon entropy (base 2) of donor labels among the $k = 50$ nearest neighbors of each cell in the integrated embedding. Higher = donors are more uniformly mixed locally = stronger batch correction.
- *Lineage silhouette:* silhouette coefficient against the author-provided lineage labels (**ClusterTop**). Higher = lineage clusters are more compact and well-separated = biological signal preserved.

Results

Integration quality (Figure 1, 2)

Method	Donor mixing entropy	Lineage silhouette
Uncorrected	1.044 +/- 0.959	0.198
Harmony	1.821 +/- 0.814	0.255
scVI	1.671 +/- 0.774	0.293

Harmony achieved the strongest donor mixing (+75% over the uncorrected baseline) and moderate lineage preservation (+29%). scVI traded a small amount of batch mixing (-8% relative to Harmony) for the cleanest lineage geometry (+48% over baseline, +15% over Harmony). The trade-off is consistent with the published benchmarks of Luecken et al. (Nat Methods, 2022): linear methods tend to over-mix while preserving less fine-grained biology, while deep generative models give more compact lineage clusters at the cost of slightly weaker batch removal.

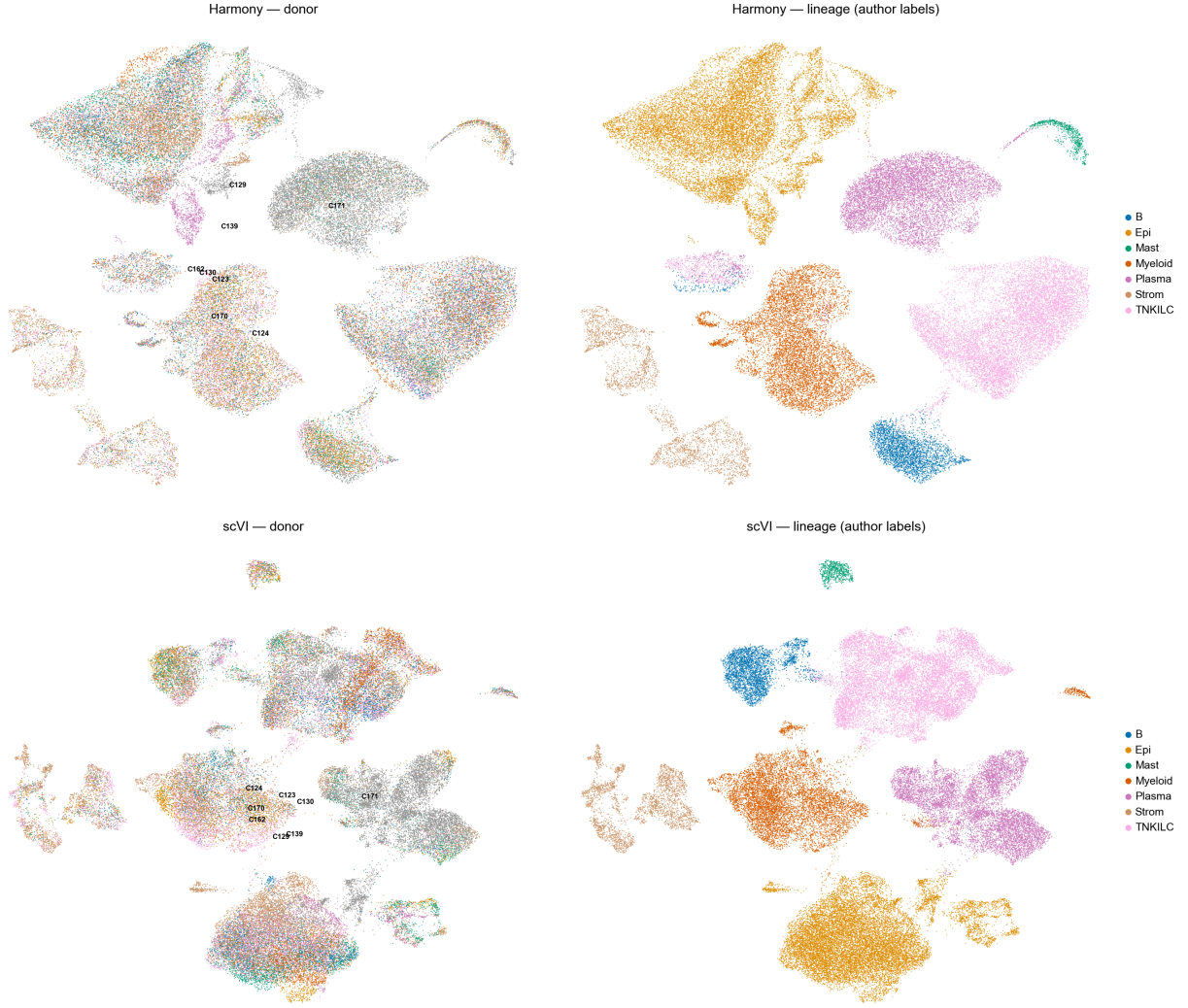


Figure 1: Harmony vs scVI integration. Top: Harmony embedding colored by donor (left) and by author-provided lineage labels (right). Bottom: scVI embedding, same colorings. Both methods recover the 7 major lineages; differences are visible in donor mixing fidelity and lineage compactness.

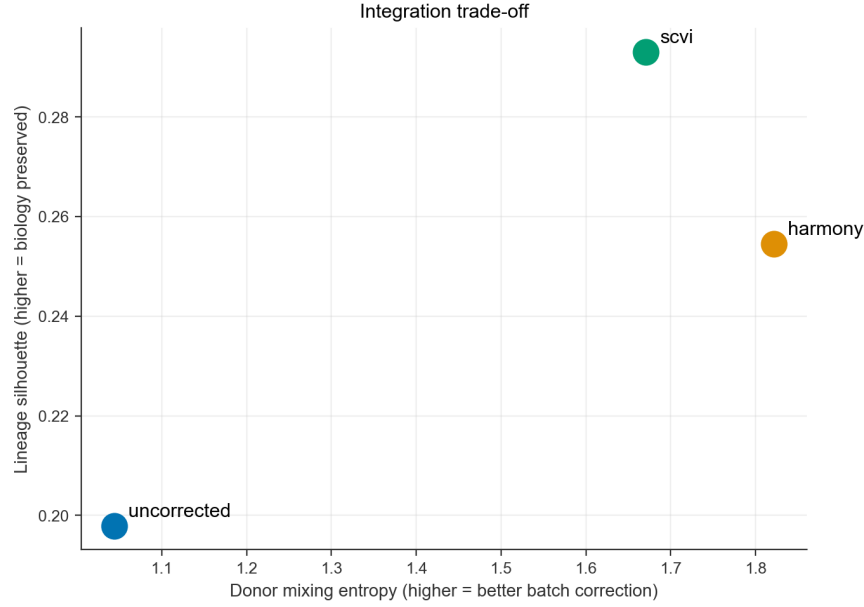


Figure 2: Integration trade-off: mixing entropy vs lineage silhouette. The uncorrected baseline sits in the low-low corner; Harmony moves furthest along the x-axis, scVI furthest along the y-axis.

Lineage composition: MMRd vs MMRp (Figure 3)

Aggregating across all eight donors, the lineage composition recapitulates the canonical immunological signature of microsatellite-unstable CRC:

- **T/NK/ILC fraction is roughly twice as high in MMRd tumors** (~0.32 vs ~0.16). This is consistent with the elevated neoantigen burden of MMRd tumors driving cytotoxic lymphocyte infiltration, and is the mechanistic basis for the FDA-approved use of immune checkpoint blockade in MSI-high CRC.
- **Plasma-cell content is markedly higher in MMRp tumors** (~0.26 vs ~0.03).
- Epithelial and myeloid fractions are similar between the two groups.

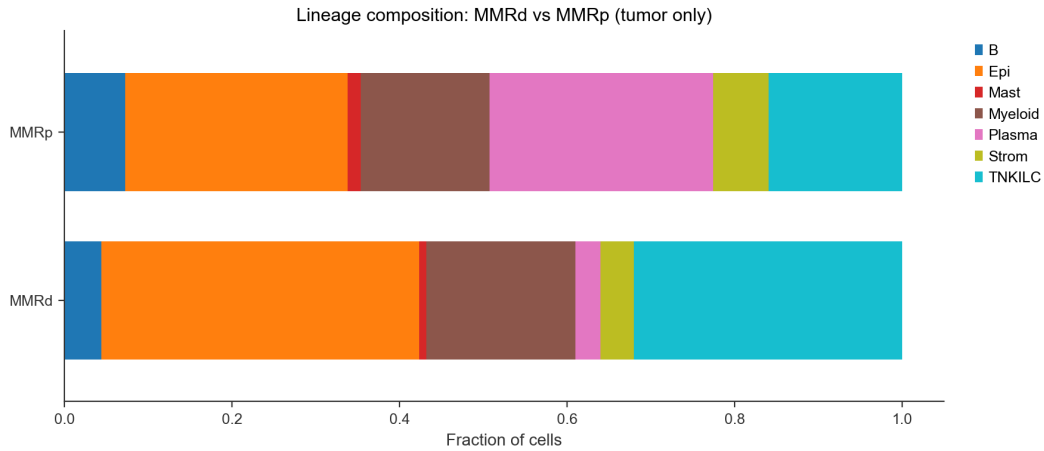


Figure 3: Per-MMR-status lineage composition. MMRd tumors are dominated by T/NK/ILC and epithelial compartments; MMRp tumors carry a substantial plasma-cell fraction.

Donor-level heterogeneity (Figure 4)

The per-donor heatmap exposes a known caveat: aggregate compositions hide within-group variance. Donor C171 (MMRp) is an outlier with > 0.45 plasma-cell fraction, single-handedly driving most of the MMRp plasma signal. Donor C170 (MMRd) skews myeloid-rich, and C123 is the most T/NK-dominated MMRd case. Any downstream claim about MMR-associated cell-state shifts at the patient level should be conditioned on this heterogeneity; an $N = 4$ comparison is not a statistical assertion of differential abundance.

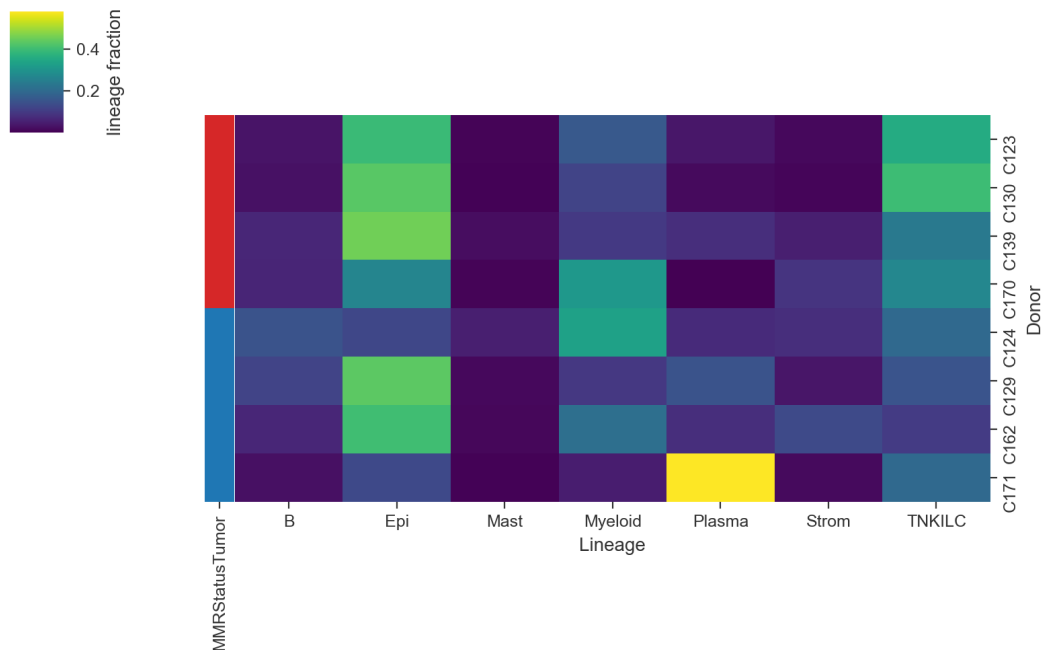


Figure 4: Per-donor lineage fractions. Red row annotation = MMRd, blue = MMRp. Note the plasma-rich C171 outlier and within-group spread.

Discussion

This re-analysis serves three purposes:

1. **Reproducibility.** The lineage structure described by the original authors is recovered with off-the-shelf Scanpy pipelines and 3,000 HVGs, without manual curation. The **ClusterTop** author labels achieve a lineage silhouette of 0.29 in scVI latent space, indicating that the modern integration toolchain agrees with the manual annotation.
2. **Method-choice guidance.** For a downstream task that requires aggressive removal of donor-level effects (e.g., joint clustering across many patients to find shared cell states), Harmony is the practical default. For tasks requiring well-separated cluster geometry (e.g., reference building, label transfer, or scArches-style projection), scVI's latent space is the better substrate. The numerical gap between them is small in absolute terms but consistent in direction.
3. **Biological signal.** The MMRd-elevated T/NK/ILC fraction is internally consistent with the known immunology of microsatellite-unstable colorectal cancer and provides a sanity check that the integration has not erased biological covariates correlated with batch.

Limitations

- $N = 4$ per MMR group is too small to support statistical claims about MMR-associated cell-type frequencies; the descriptive comparison should be read as a sanity check, not a discovery.

- The mixing-entropy and silhouette metrics are sampled (5,000 cells each); they are sufficient for ranking but not for fine-grained calibration. Full benchmarks (kBET, LISI, ASW per lineage) would refine the picture.
- scVI training was capped at 50 epochs on CPU. Longer training, particularly with a GPU, could marginally improve the integration metrics.

Reproducibility

All code, conda environment specification, and Docker recipe live in the project repository. The CELLxGENE dataset is downloaded directly from the public mirror; no manual data acquisition is required.

- Source: <https://github.com/zivanovicmkg/scrnaseq-portfolio>
- Project subdirectory: 05_colorectal_atlas/
- Environment: scanpy 1.11.5, harmonypy >= 0.0.10, scVI 1.4.2, PyTorch 2.7.1 (CPU)
- Total runtime on a 16-core / 62 GB workstation: ~45 minutes.

References

- Pelka K, et al. *Spatially organized multicellular immune hubs in human colorectal cancer*. Cell 184, 4734-4752 (2021). DOI: 10.1016/j.cell.2021.08.003
- Korsunsky I, et al. *Fast, sensitive and accurate integration of single-cell data with Harmony*. Nat Methods 16, 1289-1296 (2019).
- Lopez R, et al. *Deep generative modeling for single-cell transcriptomics*. Nat Methods 15, 1053-1058 (2018).
- Luecken MD, et al. *Benchmarking atlas-level data integration in single-cell genomics*. Nat Methods 19, 41-50 (2022).
- Wolf FA, Angerer P, Theis FJ. *SCANPY: large-scale single-cell gene expression data analysis*. Genome Biol 19, 15 (2018).